# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## 2nd Semester of B. Pharm. Examination

## University Theory Examination May 2015

## PH119 Pharmaceutical Chemistry- II

**Date: 18.05.2015, Monday** **Time: 10.00 a.m. to 1.00 p.m.** **Maximum Marks: 80**

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**Instructions:**

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat sketches wherever necessary.**

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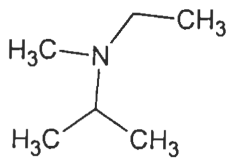
### SECTION - I

Q 1 Attempt all questions. Each question is of one mark: 20

1. Which of the following is the least acidic compound?
   [A] $CH_3COOH$
   [B] $ClCH_2COOH$
   [C] $Cl_2CHCOOH$
   [D] $Cl_3CCOOH$
2. Which of the following orbital is at the higher energy level amongst them?
   [A] 3s
   [B] 3p
   [C] 3d
   [D] 4s
3. The shape of the molecule with SP hybridization is...
   [A] Tetrahedral
   [B] Linear
   [C] Pyramidal
   [D] Trigonal
4. Which of the following has maximum bond dissociation energy?
   [A] H-H
   [B] H-F
   [C] $CH_3$-F
   [D] F-F
5. What is the name of the following compound?




   [A] Ethylmethylisopropylamine
   [B] ethylpropyl*sec*-butylamine
   [C] ethylmethylpropylamine
   [D] *sec*-butylethylpropylamin
6. Predict the product of the following reaction:
   $Et_2NH + NaNO_2/HCl \rightarrow$
   [A] $Et_3NH$ + + $Et_3N$–N=O
   [B] $Et_2N$–N=O
   [C] Et–$N_2^+$
   [D] Et–N=N–OH

7. Which of the following is a polar aprotic solvent?
   [A] DMF
   [B] Etahanol
   [C] Water
   [D] All

8. Which of the following statements regarding the E1 mechanism is wrong?
   [A] Reactions by the E1 mechanism are unimolecular in the rate-determining step.
   [B] Reactions by the E1 mechanism are generally first order.
   [C] Reactions by the E1 mechanism usually occur in one step.
   [D] Reactions by the E1 mechanism are multi-step reactions.

9. The following statement is correct for Carbanions; Except One.
   [A] Stability order of carbocation is $1^0>2^0>3^0$
   [B] Electron donating group increase the stability of Carbanion
   [C] Carbanion can act as an nucleophile
   [D] Electron withdrawing group increase the stability of Carbanion

10. Which of the following is the most stable carbocation?
   [A] $\dot{C}H_3$
   [B] ĊH2CH3
   [C] CH3ĊCH3 (with $CH_3$ on the central carbon)
   [D] CH3ĊHCH3

11. Unit of Dipole Moment is: ____________
   [A] cm-1
   [B] Poise
   [C] Debye
   [D] ppm

12. All are the examples of reactive intermediate; EXCEPT
   [A] Methane
   [B] Carbonium ion
   [C] Carbanion
   [D] Carbene

13. All statement is correct for $E_2$ reaction. Except one
   [A] Follow first order kinetic
   [B] Reactivity order is $3^0>2^0>1^0$
   [C] Always β-Hydrogen abstracted
   [D] Single step reaction

14. Orientation of Elimination reaction follows...
   [A] Markoniov's rule
   [B] Saytzeff rule
   [C] Micheal addition
   [D] Hoffmann rule

15. Which of the amine from following list can form the hydrogen bonding interaction of the type N-H---N?
   [A] Dimethylamine
   [B] Triethylamine
   [C] Pyridine
   [D] None of the above

16. Which of the following is a polar protic solvent?
   [A] Acetic acid
   [B] Etahanol
   [C] Water
   [D] All

17. Which of the following is Non Polar bond?
   [A] $H_3C—CH_3$
   [B] $H_3C—NH_2$
   [C] $H_3C—Br$
   [D] $H_3C—OH$

18. Which of the following reactive intermediate does not has any charge on it?

[A] Carbocation
[B] Carbanion
[C] Nitronium ion
[D] Carbene

19. The following statement is correct for carbocation; Except One.
[A] Stability order of carbocation is $3^0 > 2^0 > 1^0$
[B] Electron donating group increase the stability of carbocation
[C] Carbocation can act as an electrophile
[D] Electron withdrawing group increase the stability of carbocation

20. Finish the following reaction: $CH_3CH_2Li \xrightarrow{CuI} ?$
[A] $(CH_3CH_2)CuLi$
[B] $(CH_3CH_2)_2CuLi$
[C] $(CH_3CH_2)Cu$
[D] $(CH_3CH_2)_2Cu$

## SECTION – II

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| **Q 2** | **Attempt any FOUR of the following:** | |
| A | Explain positive and negative inductive effect with suitable examples. | 05 |
| B | Explain the stability of carbocation with suitable examples. | 05 |
| C | Write down various methods for preparation of alkyl halides. | 05 |
| D | Justify with suitable examples: E1 reactions are accompanied by rearrangement. | 05 |
| E | Explain various factors affecting rate of SN1 reaction. | 05 |
| F | Give explanation of Saytzeff and Markovnikov rule with suitable examples. | 05 |

## SECTION – III

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| **Q 3** | **Attempt any FOUR of the following:** | |
| A | Explain Inter molecular forces & Intra molecular forces with example. | 05 |
| B | Give explanation of $SP^3$ Hybridization. | 05 |
| C | Write a short note on: Fate and generation of Carbenes and Carbanion. | 05 |
| D | Explain mechanism of Hoffmann degradation of amides. | 05 |
| E | Explain Bond length and bond polarity with suitable examples. | 05 |
| F | Write a detailed note on mechanism of SN2 reaction. | 05 |
| **Q 4** | **Attempt any FOUR of the following:** | |
| A | Differentiate E1 & E2 reactions. | 05 |
| B | Give various methods for preparation of Alkenes. | 05 |
| C | Write a short note on Sandmeyer reaction. | 05 |
| D | Explain mechanism for halogenation reaction of alkanes. | 05 |
| E | Give any two reactions for preparation of Alkane from reduction of Carbonyl compounds. | 05 |
| F | Give IUPAC Name of following compounds: | 05 |